Arunabha Majumder

Mechanical Design Engineer | Robotic Joints, Actuators & Exoskeletons | Mechatronic Integration

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SUMMARY

Mechanical design engineer who builds exoskeletons and robotic actuators from CAD to tested hardware. Designs mechanical assemblies for robotic joints, linkages and actuator transmissions in SolidWorks (CSWA), integrates motors, encoders and sensors, and prototypes rapidly through 3D printing. Designed a hybrid-actuated shoulder exoskeleton that cut energy consumption by more than 25% (Best Paper Award, 2026). Programs hardware tests in Python and MATLAB.

KEY RESULTS

- **>25%** energy reduction from a new spring-plus-motor actuator design for a shoulder exoskeleton
- **>15%** lower user muscle effort with the motor and sensors integrated and controlled
- **<10%** RMS force-estimation error on a 5-bar linkage robot with variable-stiffness actuators

EXPERIENCE

Research Engineer – Exoskeleton & Robotic Actuator Design (PhD Project: VIEXO) | 2023 – Present
Aalborg University, Aalborg, Denmark

- Cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026) by designing the mechanical assembly of a shoulder exoskeleton joint with a new robotic actuation concept: a parallel spring working with a 6 Nm motor.
- Reduced user muscle effort by more than 15%, measured with EMG in tests with 12 participants, by integrating the motor, encoder and force sensors with the mechanical structure and implementing the control in software.
- Achieved below 10% RMS error in end-effector force estimation, validated against a force sensor, by working with the linkage kinematics (Jacobian) of a 5-bar planar parallel robot driven by variable-stiffness actuators.
- Shortened design iterations by rapidly prototyping lightweight joint and actuator parts in SolidWorks and 3D printing, and testing each version on custom test rigs.
- Delivered the integrated system by working across mechanical, electronics, control and biomechanics teams; supervised Bachelor's students (2023 – 2025).

MTech Researcher – Pneumatic Robotic Actuators (Master's Thesis) | 2020 – 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Reduced position-tracking error to 0.3–0.78%, outperforming classical PID, by designing a neural-network gain-scheduled controller in MATLAB/Simulink for a pneumatic artificial muscle actuator.
- Fabricated and assembled the actuator to an existing lab design, tested it on the lab's setup (valves, pressure sensors, DAQ) and characterised its load–displacement behaviour.

Bachelor's Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, machining, hands-on assembly and user testing; Best Major Project Award and INR 100,000 TATA Technologies Innovation Grant.
- Designed, fabricated and assembled a Go-Kart (2017) and a human-electric hybrid vehicle for Efficycle (2018).

SKILLS

- **Mechanical design & CAD:** SolidWorks (CSWA), mechanical assemblies, robotic joints, linkages, actuator transmissions
- **Mechatronic integration:** Motors, encoders, force/torque sensors, pressure sensors, pneumatic valves, DAQ, Arduino
- **Prototyping:** 3D printing (Bambu Lab, Ultimaker Cura), test rigs; machining and assembly (Bachelor's)
- **Programming & testing:** Python, MATLAB/Simulink, LabVIEW (basic) for hardware testing and data analysis
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer.
- Majumder, A., et al. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." IEEE CONECCT, 2022.

REFERENCES

Available upon request.